Speech Processing

1. a) Briefly explain the relative advantages and disadvantages of ***statistical model-based*** and ***end-to-end trained*** Automatic Speech Recognition (ASR) systems. [4]

Solution:
Statistical model-based
Pros: exploits know information structure; few parameters so can be quicker to train; explainable.
Cons: makes assumptions implicit in the statistical model (e.g. gaussian pdfs for feature distributions).
End-to-end trained
Pros: no statistical assumptions.
Cons: More parameters mean longer training, not usually explainable.

b) Consider a statistical model-based ASR system, including a trigram language model.

i) Draw a block diagram of this ASR system. [2]

ii) Using the notation

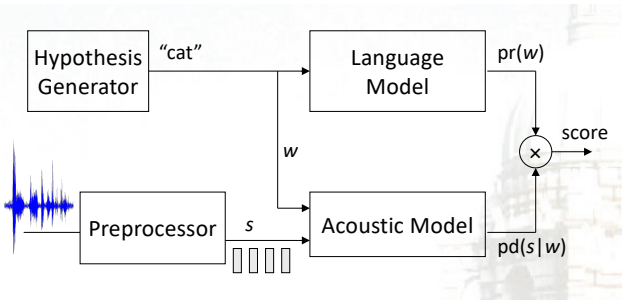
$$w[i] \text{ for } i = \ldots, -2, -1, 0, 1, 2, \ldots$$

to represent a word in a word sequence with $i = 0$ denoting the current word, write down the equation for the output of the language model. [2]

iii) Explain how the language model can be trained when the quantity of training data is limited. [3]

Solution:

(i)



(ii)

$$pr(w[0] \mid w[-2], w[-1]).$$

(iii)

Common Trigrams:

If trigram "abc" occurs n times out of a total of N trigrams "ab?" in the training data, then estimate $pr(c \mid ab)$ as n/N

Rare Trigrams:

Consider all the rare trigrams whose true probability is $0.5/N$. Some will occur once and have an overestimated probability of $1/N$. Others will not occur at all and have an underestimated probability of $0/N$.

Discounting: for trigrams with $n = 3, \ldots 6$ make the probability smaller than n/N to compensate for the overestimation;

Backing-off: for trigrams with $n = 0, 1$ or 2 use the bigram probability: estimate $pr(c \mid ab)$ as $pr(c \mid b) \times k$.

k should be chosen to ensure that

$$\sum_{all\ w} pr(w \mid ab) = 1.$$

If necessary, the bigram probabilities can be estimated from unigrams in the same way.

c) Consider a hidden Markov model-based ASR system for which a speech utterance is represented by T frames of feature vectors, $\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_T$. The utterance is compared with a Hidden Markov model having S states. The transition probability from state i to state j of the model is denoted by a_{ij} and the output probability density of frame t in state i is denoted by $d_i(\mathbf{x}_t)$.

The highest probability density that the model generates frames

$$\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_t$$

from any sequence of states for which frame 1 is in state 1 and frame t is in state s is denoted $B(t,s)$.

i) Explain how, for $t > 1$, $B(t,s)$ can be expressed in terms of $B(t-1,i)$ for $i = 1,2,\ldots,S$. Indicate the values that should be given to $B(1,i)$. [3]

ii) In a speech recognition experiment, an utterance represented by $T = 5$ frames is compared with a hidden Markov model that has 3 states. Table 1 shows the output probability density of each frame from each state of the model and Figure 1.1 shows the state diagram of the model. The figures in circles are the state numbers and the figures on arrows indicate the transition probabilities.

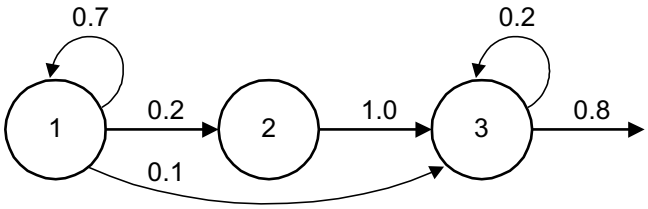
	frame				
	$\mathbf{x}_1$	$\mathbf{x}_2$	$\mathbf{x}_3$	$\mathbf{x}_4$	$\mathbf{x}_5$
state 1	0.5	0.1	0.1	0.6	0.3
state 2	0.2	0.1	0.2	0.2	0.2
state 3	0.3	0.4	0.5	0.4	0.5

Table 1

Determine the value of $B(5,3)$ and the state sequence to which it corresponds. Use six decimal places in your calculations.

[6]

Figure 1.1



Solution:

(i)
The best path for $\mathbf{x}_1,\ \mathbf{x}_2,\ \ldots,\ \mathbf{x}_t$ with t in state s must have frame $t-1$ in one of the states, denoted state i. Since the sub-path $\mathbf{x}_1,\ \mathbf{x}_2,\ \ldots,\ \mathbf{x}_{t-1}$ must also be optimum, we must have

$$B(t,s) = B(t-1,i) \times a_{is} \times d_s(\mathbf{x}_t).$$

Since $B(t,s)$ represents the probability density of the best path, we must have

$$B(t,s) = \max_{1 \le i \le S} \left(B(t-1,i) \times a_{is} \right) \times d_s(\mathbf{x}_t).$$

Since we require frame 1 to be in state 1, $B(1,s) = d_1(\mathbf{x}_1)$ if $s = 1$ and 0 otherwise.
(ii)
$B(1,1) = 0.5$
$B(2,1) = 0.5 \times 0.7 \times 0.1 = 0.035$
$B(2,2) = 0.5 \times 0.2 \times 0.1 = 0.01$
$B(2,3) = 0.5 \times 0.1 \times 0.4 = 0.02$
$B(3,1) = 0.035 \times 0.7 \times 0.1 = 0.00245$
$B(3,2) = 0.035 \times 0.2 \times 0.2 = 0.0014$
$B(3,3) = \max\{0.035 \times 0.1 = 0.0035, 0.01 \times 1 = 0.01, 0.02 \times 0.2 = 0.004\} \times 0.5 = 0.005$
$B(4,1) = 0.00245 \times 0.7 \times 0.6 = 0.001029$
$B(4,2) = 0.00245 \times 0.2 \times 0.2 = 0.000098$
$B(4,3) = \max\{0.00245 \times 0.1 = 0.000245, 0.0014 \times 1 = 0.0014, 0.005 \times 0.2 = 0.001\} \times 0.4 = 0.00056$
$B(5,3) = \max\{0.001029 \times 0.1 = 0.0001029, 0.000098 \times 1 = 0.000098, 0.00056 \times 0.2 = 0.000112\} \times 0.5 = 0.000056$
The best path is $[1,1,2,3,3]$. This may be obtained by tracing the underlined values backwards from $B(5,3)$.

2. a) Consider the simplified block diagram of a speech coder shown in Fig. 2.1.

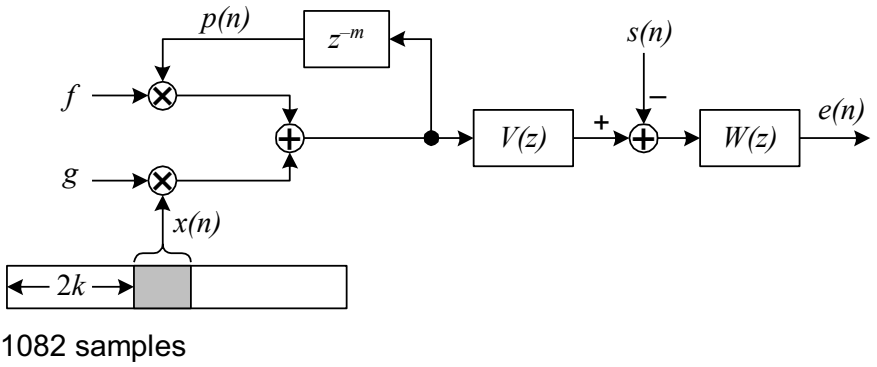


Figure 2.1: Simplified block diagram of a speech coder

i) Describe the meaning and purpose of the signals labelled $p(n)$, $x(n)$, $s(n)$ and $e(n)$. [4]

ii) Identify which elements of this coder are used to represent the periodic components of the speech excitation. [2]

iii) Explain in detail how the non-periodic components of the speech excitation signal are represented in this coder and how they are used to code a speech signal during the operation of the coder. *Hint: it is sufficient to explain what operations and functions are performed and it is not necessary to elaborate mathematically on how the calculations are performed.* [3]

Solution:
(i) p is the period excitation from the long-term predictor, x is the non-periodic excitation from the stochastic codebook, s is the original speech signal to be encoded, e is the perceptually weighted coding error to be minimized.
(ii) The periodic components are encoded using the delay z^{-m} and the gain factor f.
(iii) For each sub-frame segment (7.5 ms), after searching for the optimal LTP delay, m, and the optimal LTP gain, f, the optimal stochastic codebook entry, k, is found by exhaustive search and the corresponding optimal gain, g is determined so as to minimize the mean (sum) squared error of e in the current frame.

b) i) Consider an analogue, continuous-time speech signal $s(t)$ that is to be digitally encoded as a quantized discrete-time signal $s(n)$. Given that $p(s)$ denotes the probability density function of s, state which characteristics of $p(s)$ would make non-uniform quantization more effective than uniform quantization. Justify your answer. [3]

ii) Explain the aims of *waveform coding* and *perceptual coding*, in the context of coding speech signals. State an example of each approach and the corresponding coding gains in each case that could be typically expected. [3]

iii) A PCM-encoded speech signal $s(n)$ uses 16 bits per sample. The signal $s(n)$ is next transcoded to μ-law encoding in which S, e and m denote the sign, exponent and mantissa bits, respectively, and the quantization scheme has bin centres at

$$\pm\Big((m+16.5)2^e - 16.5\Big).$$

A particular sample of $s(n)$ has the value 66. Find the values of S, e and m in the μ-law encoding of this sample, and state the absolute value of the coding error associated with this sample.

[5]

Solution:
(i) In a non-uniform quantization scheme, the bin centres are typically chosen to match the (non-uniform) pdf of the input signal. Speech normally has a non-uniform pdf (super-Gaussian) and therefore lower quantization error can be obtained using non-uniform quantization. Examples of pdfs sometimes chosen to represent speech include the Laplacian distribution.
(ii) Perceptual coding aims to achieve good audio quality of the decoded speech whereas waveform coding aims to achieve a good match in the audio waveform. The coding gain for waveform coding, such as ADPCM is around 4, whereas for perceptual coding such as AMR it can exceed 10.
(iii) s=0, e=2, m=4 which gives the encoded value of a bin centre of 65.5. Accordingly the coding error is 0.5.

3. a) i) Let $V(z)$ denote a vocal tract filter. Give the formula for the corresponding complex cepstrum representation, and explain the advantages of the complex cepstrum for speech processing. [3]

ii) Let the unit step function be $u(n)$. Consider a signal $x(n)$ and a scalar constant a such that

$$x(n) = a^n u(n)$$

with $|a| < 1$. Find the complex cepstrum of $x(n)$. [4]

You may use the identity

$$\log(1-\alpha) = -\sum_{n=1}^{\infty} \frac{\alpha^n}{n} \text{ for } |\alpha| < 1.$$

Solution:
(i)

$$c_n = \frac{1}{2\pi} \int_{-\pi}^{\pi} \log\big(V(e^{j\omega})\big) e^{j\omega n} d\omega$$

The log function in the definition of the cepstrum, means that z-transforms that are multiplied become additive in the cepstral domain. This offers the potential to separate, for example, the voice excitation and the vocal tract transfer function.
(ii)
Take z-transform to obtain

$$X(z) = \frac{1}{1-az^{-1}}, \quad |z| > |a|.$$

Then, taking logs

$$\hat{X}(z) = \log\big(X(z)\big) = -\log(1-az^{-1}) = \sum_{n=1}^{\infty} \Big(\frac{a^n}{n}\Big) z^{-n}$$

The complex cepstrum is obtained by comparing terms in z^{-n} to give

$$\hat{x}(n) = \frac{a^n}{n} u(n-1).$$

b) Consider the mel scale mapping of frequency in Hertz given by

$$\text{mel}(f) = 2595 \log_{10}\Big(1 + \frac{f}{700}\Big).$$

i) Let

$$\Delta f_1 = 200 - 100 \text{ Hz},$$
$$\Delta f_2 = 2600 - 2500 \text{ Hz}.$$

Calculate Δm_1 and Δm_2 as the differences in mel corresponding to Δf_1 and Δf_2, respectively.

Comment on your answer. [4]

Solution:
(i)
$\Delta m_1 = 283.2 - 150.5 = 132.7$ mel
$\Delta m_2 = 1747.5 - 1712.8 = 34.7$ mel

The significant point is that the mel scale has a reduced frequency resolution at high frequencies compared to lower frequencies, matching approximately human perception.

c) The Mel-frequency cepstral coefficients can be computed using (among other functions) the FFT of a frame of speech signal samples. With the aid of a block diagram, explain in detail all the steps needed. [5]

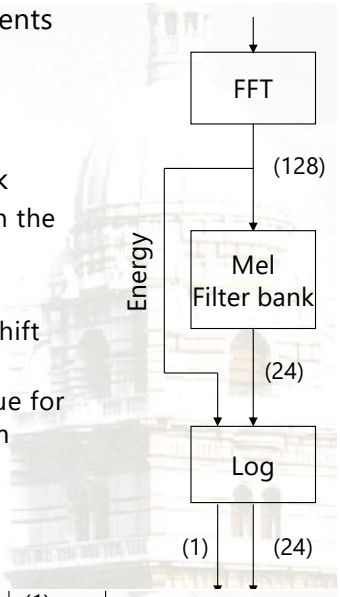
Solution:

Divide signal into overlapping 25 ms segments at 10 ms intervals
- Frame rate = 100 frames/s (Hz)

Apply Hamming window and take FFT

Smooth the spectrum with a mel filter bank
- Mel filter bank concentrates data values in the more significant part of the spectrum

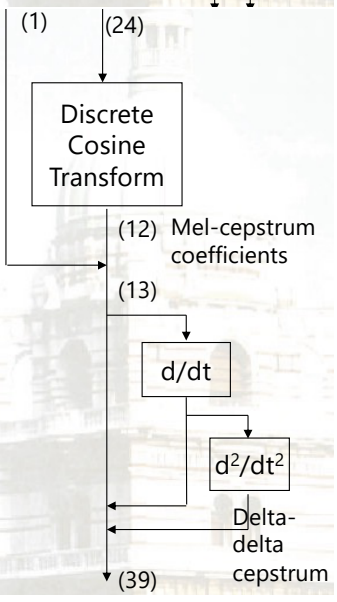
Take the log of the mel spectrum
- Variations in signal level just cause a DC shift in the log spectrum
- Gaussian approximation is more nearly true for log spectrum than for the power spectrum directly



Discrete Cosine Transform (DCT)
- reduces correlation between coefficients
- compresses information into fewer low-order coefficients
- output is the *mel-cepstrum*
- DC component is ignored to make it independent of signal level

First and Second time derivatives
- provide additional information about how the spectrum is changing with time

Result is a 39 element feature vector
- Sometimes 38 if the log energy is not included



d) A frame of speech samples is represented by a feature vector, $\mathbf{x}$, containing N Mel-frequency cepstral coefficients. The feature vector elements are known to be somewhat correlated.

Let $\mathbf{z}$ denote a vector with the same dimension as $\mathbf{x}$. Each element of $\mathbf{z}$ is an independent Gaussian distributed random variable.

It is decided to model $\mathbf{x}$ using a non-singular matrix $\mathbf{A}$ as

$$\mathbf{x} = \mathbf{A}\mathbf{z}.$$

i) Write down an expression for the covariance matrix of $\mathbf{x}$ in terms of matrix $\mathbf{A}$. [1]

ii) Hence, find an expression for the probability density function of $\mathbf{x}$. [3]

Solution:
(i)

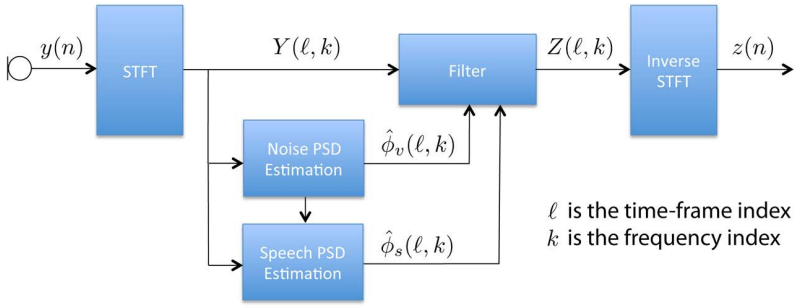
$$\begin{aligned}\mathbf{C} &= \mathrm{E}\left(\mathbf{x}\mathbf{x}^T\right) \\ &= \mathbf{A}\mathrm{E}\left(\mathbf{z}\mathbf{z}^T\right)\mathbf{A}^T = \mathbf{A}\mathbf{A}^T\end{aligned}$$

(ii)
Since $\mathbf{z} = \mathbf{A}^{-1}\mathbf{x}$ and $d\mathbf{x} = |\mathbf{A}|d\mathbf{z}$, we can write

$$\mathrm{pr}(\mathbf{X} \in \mathbf{x}, \mathbf{x} + d\mathbf{x}) = \left(2\pi\right)^{-N/2}|\mathbf{C}|^{-1/2}\exp\left(-0.5\mathbf{x}^T\mathbf{C}^{-1}\mathbf{x}\right)d\mathbf{x}.$$

4. a) Draw a labelled block diagram of a single-channel noise reduction system for speech enhancement [4]

Solution:



b) A particular single-channel noise reduction system implements

amplitude spectral subtraction.

The noisy signal in time frame l and at frequency bin k is denoted $Z(l,k)$. Describe in detail the operation of this noise reduction system and state the information needed for the system to operate correctly. [5]

Solution:
The description follows directly from the general description but employs power spectral estimation and noise power spectral variance estimation. Additional credit will be given for the depth of understanding shown, particularly concerning methods for estimation of the noise variance. For amplitude spectral subtraction, we can write

$$A_z(l,k) = A_y(l,k) - \sqrt{\hat{\phi}_v(l,k)}$$

c) The single-channel noise reduction system of part b) is formulated as

$$Y(l,k) = H(l,k)Z(l,k).$$

Determine the expression for $H(l,k)$. [4]

Solution:

$$H(l,k) = 1 - \sqrt{\frac{\hat{\phi}_v(l,k)}{|Y(l,k)|^2}}$$

This answer assumes the noisy phase spectrum is unchanged.

d) A multi-channel speech processing system, such as might be employed in a smartphone device for example, uses two closely spaced microphones. It can be assumed that both microphones receive the same speech signal $s(n)$. The speech signals $s_1(n)$ at microphone 1 and $s_2(n)$ at microphone 2 are corrupted by independent additive noise sources $e_1(n)$ and $e_2(n)$, respectively, so that

$$s_1(n) = s(n) + e_1(n)$$
$$s_2(n) = s(n) + e_2(n).$$

The noise signals have the following properties:

$$\begin{aligned}
&\mathrm{E}(e_1(n)) = \mathrm{E}(e_2(n)) = 0 \\
&\mathrm{E}(e_1^2(n)) = \sigma_1^2 \\
&\mathrm{E}(e_2^2(n)) = \sigma_2^2,
\end{aligned}$$

where $\mathrm{E}(\cdot)$ denotes the expectation operator.

A multichannel noise reduction system is now implemented to give an output

$$y(n) = ax_1(n) + (1-a)x_2(n)$$

and a is a scalar constant to be determined.

i) Determine the residual noise remaining after noise reduction, $e_3(n)$, given that

$$y(n) = s(n) + e_3(n).$$

[2]

ii) Find the value of the constant a that minimizes the residual noise power of $e_3(n)$. [5]

Solution:
(i)

$$e_3(n) = ae_1(n) + (1-a)e_2(n)$$

(ii)
The variance of the combined noise is found as

$$\begin{aligned}
\sigma_3^2 &= \mathrm{E}\Big(\big(ae_1(n) + (1-a)e_2(n)\big)^2 \Big) \\
&= a^2\mathrm{E}\Big(e_1^2(n)\Big) + (1-a)^2\mathrm{E}\Big(e_2^2(n)\Big) \\
&= a\sigma_1^2 + (1-a)^2\sigma_2^2
\end{aligned}$$

The minimization is obtained by differentiation w.r.t. a and setting to zero, giving

$$\begin{aligned}
\frac{\partial \sigma_3^2}{\partial a} &= 2a\sigma_1^2 - 2(1-a)\sigma_2^2 = 0 \\
&\Longrightarrow 2a\sigma_1^2 + 2a\sigma_2^2 = 2\sigma_2^2 \\
&\Longrightarrow a = \frac{\sigma_2^2}{\sigma_1^2 + \sigma_2^2}.
\end{aligned}$$